

AI & MACHINE LEARNING — TASK 1

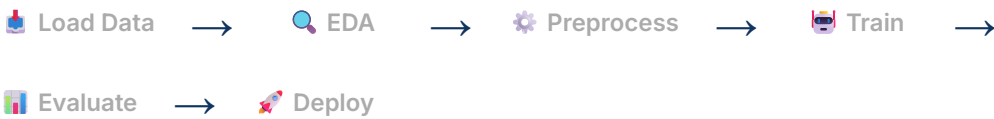
Linear Regression: House Price Predictor

Build & Evaluate a Machine Learning Model on the California Housing Dataset

Dataset	Algorithm	Samples
California Housing (scikit-learn)	Linear Regression (OLS)	20,640 instances
Features	Environment	
8 numerical features	Urban vs. Suburban	

Objective

This project introduces the complete **Machine Learning lifecycle** — from raw data exploration to model deployment — using a real-world housing dataset. We train a **Linear Regression** model to predict median house values across California block groups, evaluate it using standard regression metrics, and provide a deployable prediction interface.



Dataset Overview

The **California Housing Dataset** is derived from the 1990 U.S. Census. Each row represents a *block group* — the smallest geographical unit for which the Census Bureau publishes sample data. The dataset contains **20,640 samples** with no missing values.

Feature	Description	Mean	Std Dev
MedInc	Median household income (×\$10,000)	3.8707	1.8998

Feature	Description	Mean	Std Dev
HouseAge	Median house age (years)	28.64	12.59
AveRooms	Average rooms per household	5.43	2.47
AveBedrms	Average bedrooms per household	1.10	0.47
Population	Block group population	1425.5	1132.5
AveOccup	Average household members	3.07	10.39
Latitude	Block group latitude	35.63	2.14
Longitude	Block group longitude	-119.57	2.00
MedHouseVal	Target: Median house value (×\$100,000)	2.0686	1.1540

Exploratory Data Analysis (EDA)

Data Quality



No missing values were found across all 20,640 rows and 9 columns. The dataset is clean and requires no imputation.

Target Variable Distribution

The target variable **MedHouseVal** has a mean of **\$206,860** and a median of **\$179,700**. The distribution is *right-skewed* with a hard cap at \$500,001 (values clipped at 5.0 in the dataset), creating a visible spike at the right tail. This clipping may slightly limit model accuracy for high-value properties.

Feature Correlations with Target

Feature	Correlation with MedHouseVal	Direction
MedInc	0.6882	Strong Positive ↑
AveRooms	0.1512	Weak Positive ↑
HouseAge	0.1059	Weak Positive ↑
AveOccup	-0.0237	Very Weak Negative ↓
Population	-0.0248	Very Weak Negative ↓

Feature	Correlation with MedHouseVal	Direction
Longitude	-0.0455	Weak Negative ↓
Latitude	-0.1446	Weak Negative ↓
AveBedrms	-0.0467	Weak Negative ↓

💡 **Key Insight:** *MedInc* (median income) is by far the strongest predictor of house value, with a correlation of **0.69**. Geographic features (Latitude, Longitude) also carry significant predictive power, capturing the coastal premium effect in California real estate.

Methodology

Preprocessing Pipeline

1. **Feature/Target Separation:** Split dataset into X (8 features) and y (MedHouseVal)
2. **Train/Test Split:** 80% training (16,512 samples) / 20% test (4,128 samples) with random_state=42
3. **Feature Scaling:** Applied StandardScaler to normalize all features to zero mean and unit variance — prevents features with large magnitudes (e.g., Population) from dominating the model

```
# Preprocessing pipeline X_train, X_test, y_train, y_test = train_test_split( X,
y, test_size=0.20, random_state=42 ) scaler = StandardScaler() X_train_scaled =
scaler.fit_transform(X_train) # fit on train only X_test_scaled =
scaler.transform(X_test) # no data leakage # Model training model =
LinearRegression() model.fit(X_train_scaled, y_train)
```

Model: Linear Regression

Linear Regression fits the following equation using **Ordinary Least Squares (OLS)**, which minimizes the sum of squared residuals:

$$\hat{y} = \beta_0 + \beta_1 \cdot \text{MedInc} + \beta_2 \cdot \text{HouseAge} + \beta_3 \cdot \text{AveRooms} + \beta_4 \cdot \text{AveBedrms} + \beta_5 \cdot \text{Population} + \beta_6 \cdot \text{AveOccup} + \beta_7 \cdot \text{Latitude} + \beta_8 \cdot \text{Longitude}$$

Learned Coefficients

Feature	Coefficient (β)	Interpretation
MedInc	+0.8530	Highest positive impact on price
AveRooms	+0.1214	More rooms → higher price
HouseAge	+0.1228	Older houses → slightly higher price
AveBedrms	-0.0541	More bedrooms/house → slightly lower
Population	-0.0052	Denser areas → very slightly lower
AveOccup	-0.0390	More occupants → slightly lower
Latitude	-0.8975	Moving north → lower price (coastal effect)

Feature	Coefficient (β)	Interpretation
Longitude	-0.8679	Moving inland → lower price

Results & Evaluation

MAE — TEST SET 0.5332 Avg error ≈ \$53,320	RMSE — TEST SET 0.7456 ~\$74,560 root error	R² — TEST SET 0.5757 57.6% variance explained
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Metric	Formula	Training	Test	Interpretation
MAE	$\text{mean}(y - \hat{y})$	0.5274	0.5332	Avg error of ~\$53K per prediction
RMSE	$\sqrt{\text{mean}((y - \hat{y})^2)}$	0.7277	0.7456	Penalizes large errors more
R ² Score	$1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}}$	0.6027	0.5757	Model explains 57.6% of variance

 **Analysis:** The train and test metrics are very close (R²: 0.603 vs 0.576), indicating that the model is **not overfitting**. The slight gap is expected. For a simple linear model on this dataset, an R² of ~0.58 is the typical baseline — the relationship between features and price is not perfectly linear, which limits the model's ceiling.

Improvement Ideas

1. Feature Engineering Create derived features: rooms_per_person, income_per_room. Non-linear combinations can capture patterns that raw features miss.	2. Outlier Treatment Cap extreme values in AveRooms, AveBedrms, Population using IQR or percentile clipping to reduce noise.
3. Advanced Models Try Ridge/Lasso regression (regularization), Random Forest, or XGBoost — expected R² improvement to 0.80+.	4. Polynomial Features Add degree-2 interaction terms using PolynomialFeatures to capture non-linear relationships between features.

5. Log Transform Target

Apply `np.log(y)` before training to reduce right skew in the target distribution and potentially improve predictions.

6. Cross-Validation

Use `KFold` or `cross_val_score` with `k=5` for more robust performance estimation instead of a single train/test split.

Deliverables

- **Notebook** `task1_ml_linear_regression.ipynb` — Complete ML workflow with code, plots, and comments
 - **Report** `report.html / report.pdf` — This document (2–4 pages)
 - **Script** `predict_ui.py` — Interactive CLI predictor using the saved model
 - **Model** `house_price_model.pkl` — Saved `LinearRegression` + `StandardScaler`
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Maincraft Internship — AI & Machine Learning

Task 1: Linear Regression | California Housing Dataset

Model: `LinearRegression` (scikit-learn)

Test R^2 Score: 0.5757